# CSC1006 - Final Exam Sample

## Instructions

1. The Exam consists of 5 pages (including this one). The notations and definitions are those used in class.
2. This is a **closed-book exam**.
3. The aid of electronic devices **of any form** is strictly prohibited and will be regarded as cheating.
4. Students who are caught cheating will have their Final-Term Exam collected at once and will be reported to the Authorities in charge.
5. Utilise a clear and legible handwriting to have your work properly graded.
6. You have exactly 90 minutes to complete the Final Exam.

# 1 Multiple Selection ($3 \times 10 = 30$ pts)

1. Which of the following statement is true?
   - A. Visualization is an essential part in machine learning and deep learning
   - B. Machine learning and deep learning have found wide-spread application in science and engineering
   - C. Deep learning learns from data with deep neural networks.
   - D. All of the above
2. You're fitting a multi-variable linear regression model to predict house prices using two features: size (sq ft) and age (years). Which code snippet correctly predicts price for 1300 sqft, 3-year-old home.
   - A. `pred = model.predict(1300)`
   - B. `pred = model.predict([[1300, 3]])`
   - C. `pred = model.predict([[1300], [3]])`
   - D. `pred = model.predict(1300, 3)`
3. Why does the binary cross entropy loss penalize incorrect predictions sharply?
   - A. The logarithmic terms cause the loss to grow slowly as $\hat{y}$ approaches 0 or 1 incorrectly
   - B. The logarithmic terms cause the loss to approach infinity as $\hat{y}$ approaches the wrong extreme
   - C. The loss is capped at a fixed value for incorrect predictions
   - D. The loss is linear in the difference between $\hat{y}$ and $y$
4. What is a likely consequence of overfitting in a machine learning model?
   - A. The model captures the training data's patterns too closely, including noise, leading to poor generalization to unseen data.
   - B. The model generalizes well to new data but performs poorly on the training set.
   - C. The model is too simple to learn meaningful patterns from the training data.
   - D. The model achieves high accuracy on both training and validation sets consistently.
5. You are preparing a dataset for a machine learning task and need to split it into training and test sets. The following code is proposed:

```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

   You notice the model's performance varies significantly across runs. What modification would improve consistency?
   - A. Adjust the model hyper-parameters.
   - B. Increase `test_size` to 0.5 for a larger test set.
   - C. Shuffle `X` and `y` manually before splitting.
   - D. Add `random_state=42` to ensure reproducible splits.
6. In a pair of images captured by a stereo camera, what kind of information could not be acquired or computed?
   - A. Color.
   - B. Brightness.
   - C. Temperature.
   - D. Distance.
7. What does a Neural Radience Field (NeRF) model fundamentally learn to represent??
   - A. A 3D mesh with texture maps extracted from images.
   - B. Relationship between 3D position and viewing direction to pixel color and density.
   - C. A dense and robust 3D point cloud reconstructed via depth sensors.
   - D. A continuous relationship of different camera poses among a lot of images.
8. During rendering, NeRF accumulates color along a camera ray using volume rendering. Which of the following best describes the role of the density ($\sigma$) output?

A. It controls the learning rate of the network during training.
B. It defines the resolution of pixels in the whole image.
C. It depicts the likelihood that the light ray is affected by its position.
D. It defines the lateral distance between sampled points along the light ray.

9. To use Q-learning method to achieve reinforcement learning, what kind of the following information is needed?

   A. Learning rate.
   B. Discount factor.
   C. Bellman Equation.
   D. All of the above

10. Which of the following robots does not belong to the type of mobile robots?

    A. Floor cleaning robot.
    B. Industrial arm.
    C. Autonomous car.
    D. Four-legged robot.

## 2 ML vs DL (10 pts)

Classify each attribute below as more characteristic of traditional machine learning or deep learning.

1. Needs very large datasets (typically over a million samples) DL
2. Requires manual feature extraction and engineering ML
3. Employs deep neural network DL
4. Works well with moderate-sized, well-structured tabular data ML
5. Often uses `sklearn` for implementation ML
6. Dominates computer vision tasks such as facial recognition DL
7. Often uses decision trees ML
8. Dominates natural language processing such as language translation DL
9. Often uses `PyTorch` or `TensorFlow` for implementation DL
10. Use back-propagation to update the parameters in the model DL

## 3 Classification Loss Function (8 pts)

The binary cross entropy loss function is frequently used in binary classification problems. For a given prediction $\hat{y}$, whose true value is $y$, the binary cross entropy loss function is implemented as

```python
def binary_cross_entropy(y, y_hat):
    return -y * np.log(y_hat) - (1 - y) * np.log(1 - y_hat)
```

1. (2 pts) The function will fail with a logarithm error if $\hat{y} = 0$ or $\hat{y} = 1$. Explain why.
2. (6 pts) Explain why minimizing this loss function will drive $\hat{y}$ towards $y$.

## 4 Linear Regression and Logistic Regression (16 pts)

**Background:** In a pharmaceutical study, researchers investigate the effect of drug dose (usage amount) $x$ (in mg) on two outcomes for the same set of patients:

- $y$: reduction in blood pressure (mmHg) after 6 hours (continuous, positive means decrease).
- $z$: occurrence of side effects (binary value: $z = 1$ if side effects occur, $z = 0$ otherwise).

**Data** from 6 patients:

| Dose $x$ (mg) | BP reduction $y$ (mmHg) | Side effect $z$ |
|---|---|---|
| 10 | 1.5 | 1 |
| 20 | 3.0 | 1 |
| 30 | 5.0 | 1 |
| 40 | 6.5 | 0 |
| 50 | 8.0 | 0 |
| 60 | 9.5 | 0 |

You will build a prediction model for $y$ and $z$ separately.

**Questions:**

1. **(4 pts)** Which type of regression (linear or logistic) is more appropriate for predicting $y$ and $z$, separately? and why? If we directly fit a linear regression model $z = \beta_0 + \beta_1 x$ using the binary side-effect labels, what problems would arise?
2. **(4 pts)** For the linear regression model $y = \beta_0 + \beta_1 x$, what's the loss function to fit the model? Assuming current initialized parameters $\beta_0 = 1.0$ and $\beta_1 = 0.1$ for the the above linear regression model, calculate the loss using the data in the table.
3. **(3 pts)** The logistic regression model for side effects can be written as

$$P(z = 1 \mid x) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}.$$

   Explain why logistic regression uses the sigmoid function ($\sigma(\#) = \frac{1}{1+e^{-\#}}$) as the link function. How does its output differ from that of linear regression?
4. **(5 pts)** Write the name and formula of the loss function commonly used for logistic regression. Assuming the current parameters $w_0 = 1.0$ and $w_1 = -0.2$ for the above logistic regression model, compute the predicted probability $\hat{p}_i$ for the first patient and also compute the loss for this first patient (you don't need to calculate out the exponentiations, which in your final answer you can just write directly as exp(...) or $e^{(\cdots)}$ with correct numbers inside the bracket inside your formula).

## 5 Iris Classification (16 pts)

```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Load and split data
X, y = load_iris(return_X_y=True)
y = y == 1  # Binary classification: Versicolour (1) or not (0)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# KNN model (k=3)
knn = _____________(n_neighbors=3)
knn.fit(X_train, y_train)
knn_pred = knn.predict(X_test)

# Logistic Regression model
lr = _____________()
lr.fit(X_train, y_train)
lr_pred = lr.predict(X_test)
                                y_test    knn_pred
# Evaluate both models          y_test    lr_pred
def evaluate_model(name, y_true, y_pred):
    print(f"\n{name} Performance:")
    print(f"Accuracy: {accuracy_score(y_true, y_pred):.3f}")
```

```
26     print(f"Precision: {precision_score(y_true, y_pred):.3f}")
27     print(f"Recall: {recall_score(y_true, y_pred):.3f}")
28     print(f"F1 Score: {f1_score(y_true, y_pred):.3f}")
29
30 evaluate_model("KNN (k=3)", ______, ________)
31 evaluate_model("Logistic Regression", ______, _______)
```

Read the code and answer the following questions

1. (2 pts) Four metrics are imported in line 5 and then used to evaluated the model. What is the limitation of using only one metric such as the accuracy?
2. (6 pts) Fill in the blanks in lines 13, 18, 30, and 31.
3. (4 pts) Explain what `n_neighbors=3` means in the $k$-NN algorithm's decision process. Describe how increasing or decreasing `n_neighbors` would affect its susceptibility to overfitting/underfitting.
4. (4 pts) Provide step-by-step instructions (in pseudocode or clear English) for finding out the best `n_neighbors` for the dataset.

# 6 Neural Network (15 pts)

Consider a part of the output layer in the neural network with the activation function and the loss function given in Fig. 1. Now we have the data with $h_1 = 0.8$, $h_2 = 1$ and the expected output $y = 0.8$. If we assume that the learning rate is $\lambda = 0.2$, please use the gradient descent model $w_i \leftarrow w_i - \lambda \frac{\partial L}{\partial w_i}$ to calculated the updated weights using backpropagation.

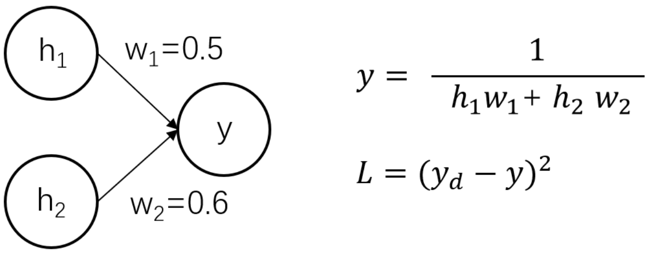


Figure 1: The (partial) neural network structure with its settings.

# 7 Robot Motion Control (5 pts)

simple
stable
real-time perf

lack adaptability

Please briefly explain the advantages and disadvantages of traditional robot motion controllers (like the proportional controller) and reinforcement learning control, respectively.

↳ adaptable in complex environment
high comp. cost
Unstable while training